

Ethics in Computational Social Science

Based on Salganik, Bit by Bit: Social Research in the Digital Age — Chapter 6

Why Ethics?

The digital age creates new opportunities — and new risks.

The Harm Risk

Some researchers have been accused of violating privacy or running unethical experiments on people without their knowledge.

The Chilling Effect

Ethical uncertainty also stops important, ethical research from happening — a less visible but equally costly problem.

Goal: develop shared norms so research can be both responsible and beneficial.

Three Cautionary Tales

01

Emotional Contagion (Facebook, 2014)

Facebook manipulated 700,000 users' news feeds to study emotional spread — without explicit consent. Sparked a massive public backlash about whether a ToS click equals research consent.

02

Tastes, Ties & Time (Harvard, 2006)

Researchers scraped and released a 'anonymized' college Facebook dataset. It was later re-identified by combining with other data — showing anonymization is far harder than it appears.

03

Encore (Internet Censorship Study)

A tool embedded in web pages quietly used visitors' browsers to probe censored sites — without their awareness. Participants in authoritarian countries could have faced serious legal risk.

Discussion: What do these cases have in common?

Why Digital Is Different

Researchers can now observe and experiment on people at massive scale — without consent, and without their awareness.



Power outpaces norms

Our technological capacity to collect and analyze data has grown faster than our ethical rules, laws, and social norms.



Old rules don't fit

Traditional IRB rules were designed for lab experiments and paper surveys — they don't map cleanly onto scraping tweets, platform A/B tests, or phone records.



The result: genuine uncertainty

Researchers face real dilemmas, not just rule-breaking. Even well-meaning researchers can cause harm without knowing it.

Three Approaches to Ethics

Rules-Based

(Social Scientists / IRB)

Follow the checklist, get IRB approval, done.

Problem: rules lag behind technology and can't cover every new situation.

Ad Hoc

(Data Scientists / Engineers)

No systematic training in research ethics — decisions made case by case with no principled framework.

Problem: arbitrary and inconsistent.

Principles-Based

(Salganik's Recommendation)

Go beyond rules to the underlying ethical principles. Helps when rules haven't been written yet, and helps researchers explain their reasoning.

This is what we adopt.

Four Core Principles

From the Belmont Report & Menlo Report

1

Respect for Persons

Treat people as autonomous agents. Requires informed consent. Extra care for vulnerable populations.

2

Beneficence

Maximize benefits, minimize harms. Covers emotional, reputational, financial and legal harm — not just physical.

3

Justice

Who bears the burden of research? Who gets the benefits? Avoid exploiting vulnerable or marginalized communities.

4

Respect for Law & Public Interest

Follow terms of service and law. Go beyond legal compliance — be transparent and consider broader societal impact.

Two Ethical Frameworks



Consequentialism

Focus: Ends matter

The right action is the one that produces the best outcomes. Harms may be justified by sufficiently large benefits.

Example: Experimenting on users without consent could be justified if it prevents serious societal harm at scale.

Risk: can rationalize almost any means if the end seems good enough.



Deontology

Focus: Means matter

Some actions are intrinsically right or wrong regardless of consequences. People must not be treated merely as instruments.

Example: Deceiving participants is wrong even if the study produces valuable knowledge.

Risk: can be too rigid when flexibility might produce much greater good.

Heuristic: If acceptable under BOTH frameworks → proceed. Only one → proceed cautiously. Neither → stop.

Four Hard Problems in Practice

1. Informed Consent

Not always required, but default to seeking it. Clicking ToS ≠ consent. Waiver may apply if: minimal risk, impractical, study couldn't otherwise happen.

2. Informational Risk

'Anonymous' data often isn't. Re-identification attacks are real. Risk isn't just to individuals — data can stigmatize entire communities.

3. Privacy

'Public' data ≠ fair game. Ask: would participants expect this use? Context matters — a public tweet ≠ consent to a mental-health dossier.

4. Uncertainty

You can't always know harms in advance. Don't let uncertainty paralyze you — use principles to reason through it, consult affected communities.

Three Practical Tips



01

The IRB is a floor, not a ceiling

Getting IRB approval clears a minimum regulatory bar — it doesn't mean your research is fully ethical. Your obligation goes beyond compliance.



02

Put yourself in everyone else's shoes

Ask: "How would participants react if they found out?" Would this look acceptable on the front page of a newspaper? Think about all subgroups, not just the average participant.



03

Think of ethics as continuous, not a checkbox

Ethical duties don't end when data collection begins. They apply throughout: storage, analysis, publication, and public communication. New harms can emerge at each stage.

Group Exercise



Scenario

You want to study online radicalization by scraping public posts from an extremist forum. Users are unaware they are being studied. Some posts could be used to re-identify individuals. The research could inform counter-terrorism policy and potentially save lives.

Discuss:

1. Which of the four principles does this raise?
2. How do consequentialism and deontology pull in different directions here?
3. What would you do, and what would you change about the research design?

Key Takeaways



Rules alone are insufficient — adopt a principles-based approach to navigate situations where rules haven't been written yet.



The four principles (Respect for Persons, Beneficence, Justice, Respect for Law) give you a shared language for ethical reasoning.



'Public data' and 'just observing' don't automatically make research ethical — context and power asymmetries matter.



Use both ethical frameworks: consequentialism (ends) and deontology (means) as complementary lenses, not competing doctrines.



Ethics is continuous — build it into every stage of your research, from design through data storage, analysis, and publication.